INTRODUCTION TO DATA ANALYTICS | MCA AI ML

Healthcare Analytics And Patient Intelligence

Excel, Power BI aur Jupyter Notebook dashboards ke saath 1,000 patients ka analysis

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Introduction aur Objective

Introduction

Hospital roz bahut data banata hai: patients, ward, bimari, doctor aur billing. Agar ye data sahi se dekha jaye to hospital ko behtar faisle lene me madad milti hai.

Is project me hospital ka data Excel se liya gaya aur Excel, Power BI aur Jupyter Notebook me dashboard bana kar analyze kiya gaya.

Objectives

- Hospital operations (ward, stay, admission) samajhna
- Patients ki age, gender aur city ka pattern dekhna
- Bimariyon ka count aur severity dekhna
- Doctors ki performance compare karna
- Cost aur revenue ka analysis karna
- Raw data ko visual insights me badalna

Data Source aur Tools

5 Excel files (har file me Raw Data + summary sheets)

File	Kya hai
1. Hospital Operation	Admission, discharge, stay, ward
2. Patient Analytics	Age, gender, city
3. Disease Analytics	Bimari, severity, age group
4. Doctor Performance	Doctor, bimari, treatment cost
5. Cost and Revenue	Cost, ward, payment mode

Tools

- **Microsoft Excel**: data, Power Query, Pivot Tables, Excel dashboard
- **Power BI**: Major Dashboard (slicers ke saath)
- **Python / Jupyter Notebook**: pandas + matplotlib dashboards

> Sabhi files me Patient ID common hai (P0001 se P1000). Dataset synthetic (generated) hai, real patient records nahi.

Data Overview

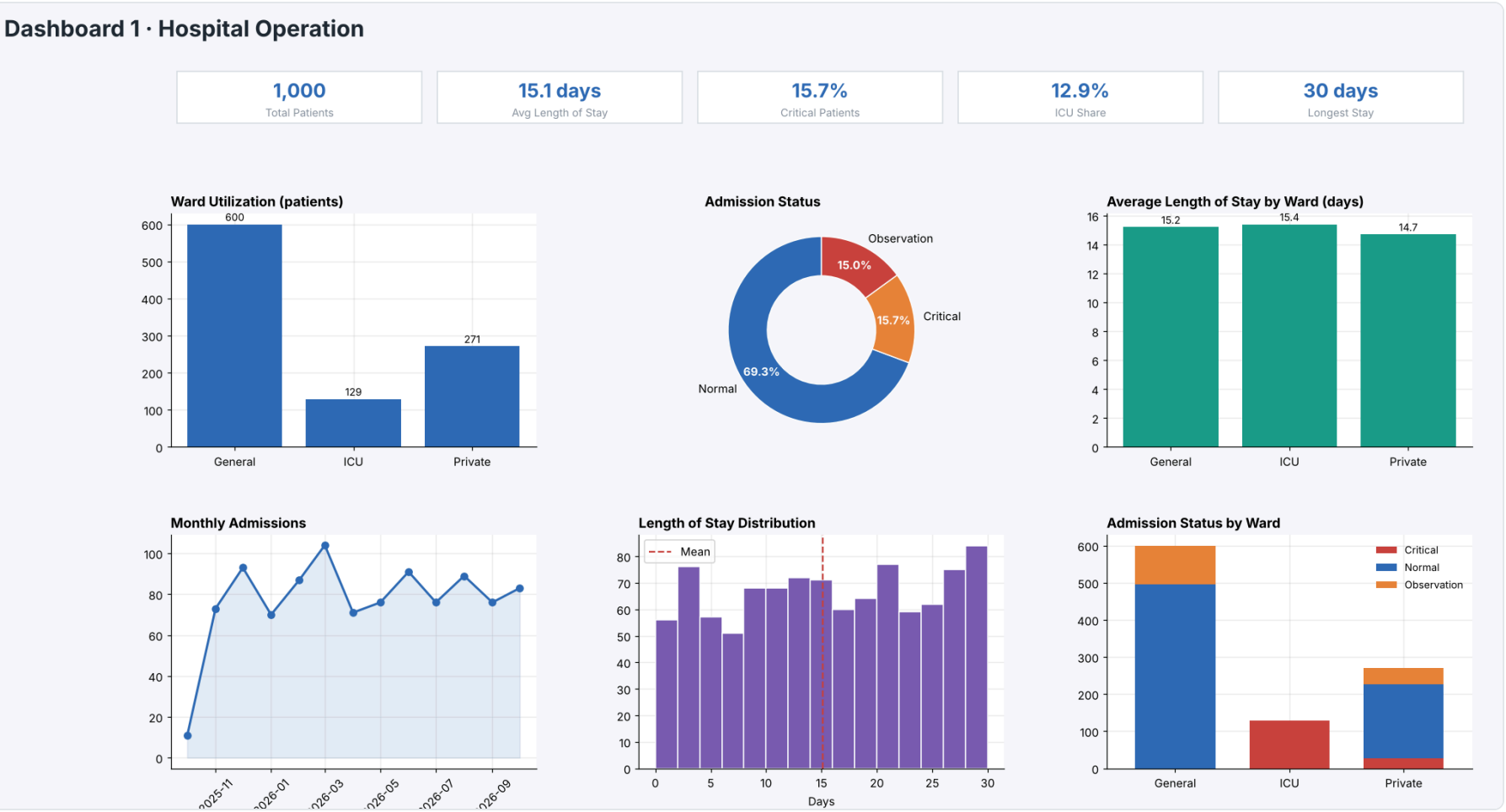
Dataset kitna bada hai

1,000 Total patients	**₹39.42 Cr** Total revenue	**₹3.94 Lakh** Avg cost per patient	**15.1 din** Avg length of stay
10 Bimariyan	**6** Doctors	**8** Cities	**0** Missing values

Data period: 28 Oct 2025 se 28 Oct 2026 (admission dates). Teeno wards: General, Private, ICU.

Hospital Operations

Dashboard 1 (Jupyter Notebook)

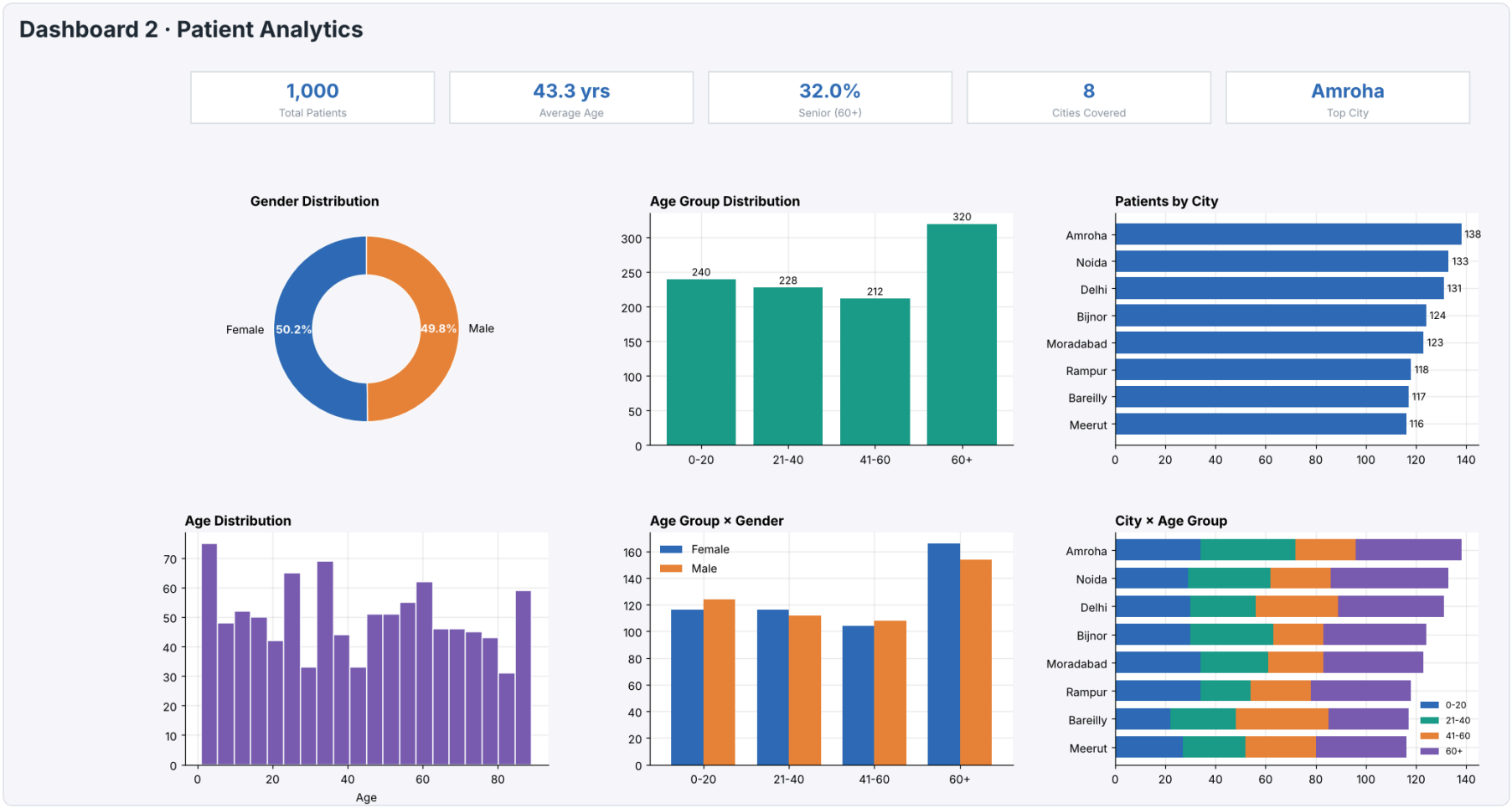


Key points

- General ward me sabse zyada 600 patients (60%), ICU me 129 (12.9%)
- 69.3% patients Normal, 15.7% Critical, 15.0% Observation
- Avg stay lagbhag same: General 15.2, ICU 15.4, Private 14.7 din
- ICU me sab patients Critical hain

Patient Analytics

Dashboard 2 (Jupyter Notebook)

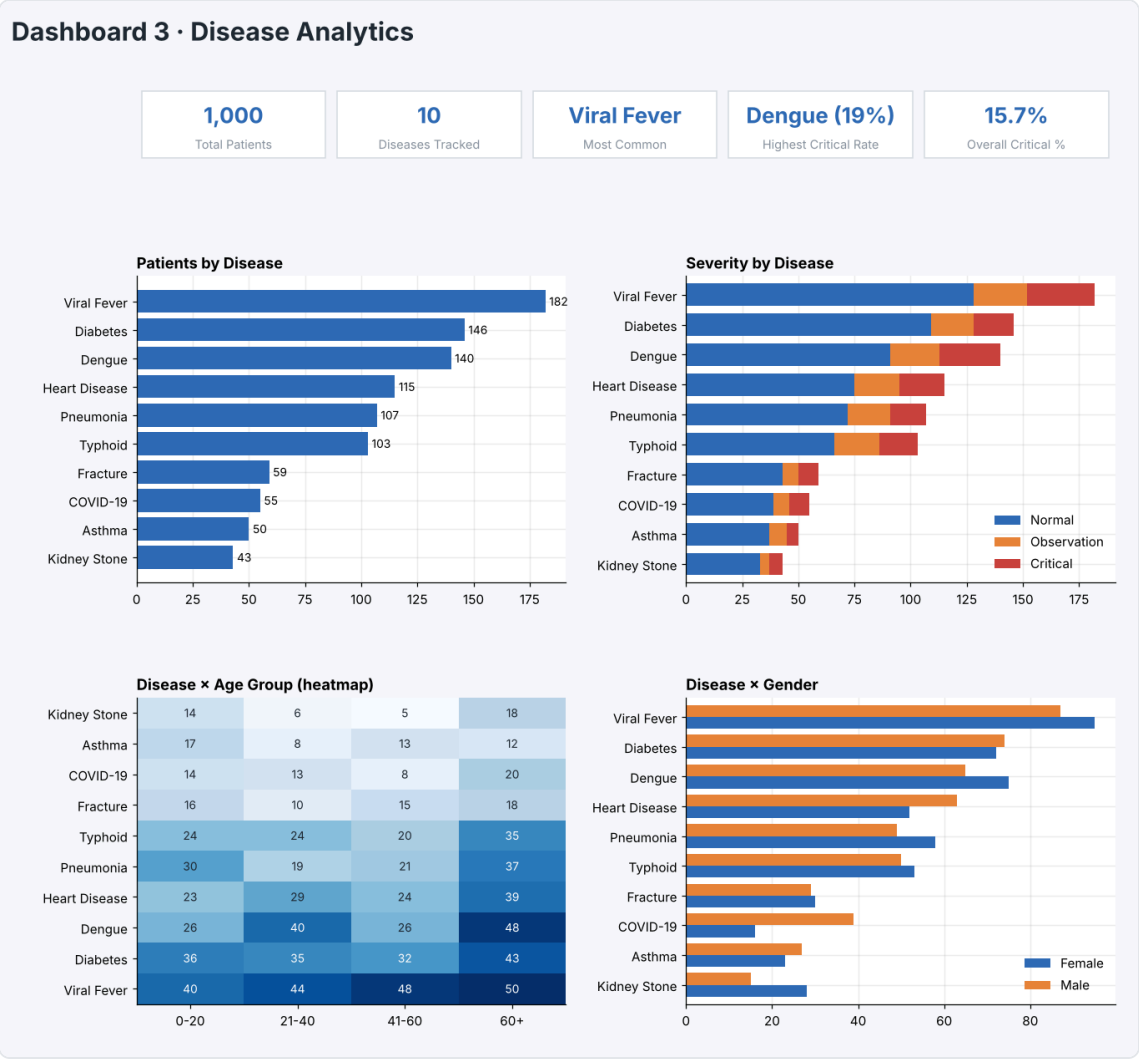


Key points

- Gender balanced: Female 502, Male 498
- Avg age 43.3 saal; 60+ group sabse bada: 320 (32%)
- Amroha (138), Noida (133), Delhi (131) se sabse zyada patients

Disease Analytics

Dashboard 3 (Jupyter Notebook)



Key points

- Viral Fever sabse common (182), phir Diabetes (146), Dengue (140)
- Dengue me critical share sabse zyada: 19.3%
- Asthma me sabse kam: 10%
- Kidney Stone sabse kam: 43 patients

Doctor Performance

Dashboard 4 (Jupyter Notebook)



Key points

- Dr. Singh (Pulmo): sabse zyada revenue ₹8.08 Cr, avg ₹4.87 Lakh per patient
- Dr. Verma (Physician): sabse zyada patients (189), par avg sabse kam ₹3.18 Lakh
- Dr. Khan: 174 patients, revenue ₹5.68 Cr

Cost aur Revenue

Dashboard 5 (Jupyter Notebook)

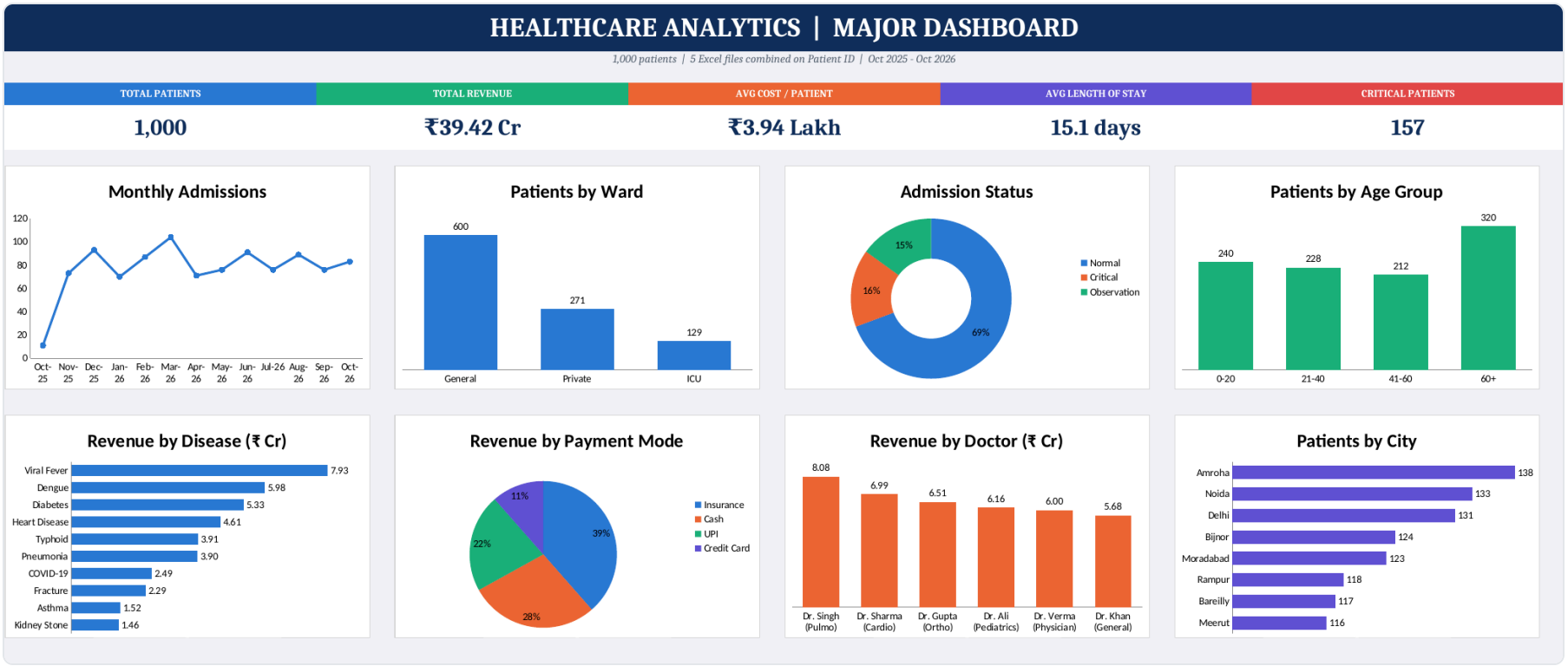


Key points

- Total revenue ₹39.42 Cr, avg bill ₹3.94 Lakh
- Insurance sabse bada payment mode: ₹15.20 Cr (38.6%)
- ICU se ₹25.20 Cr (63.9%) revenue, jabki ICU me sirf 12.9% patients
- Viral Fever se sabse zyada revenue: ₹7.93 Cr

Excel Major Dashboard

Microsoft Excel me bana Major Dashboard

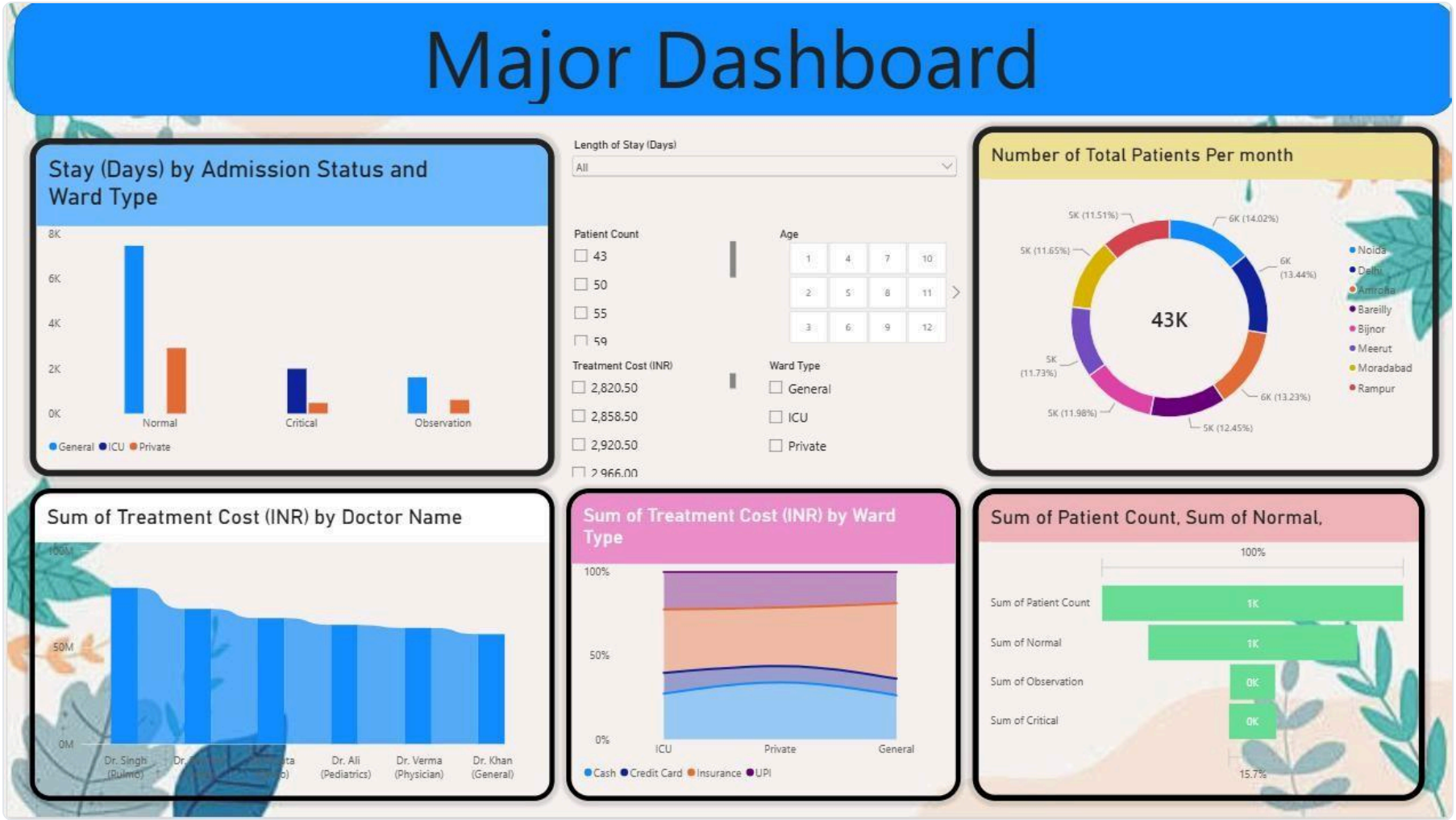


Key points

- 5 Excel dashboards ka combined Major Dashboard
- Master Data sheet me 5 files Patient ID par merge
- KPIs aur 8 charts formulas se live hain
- Revenue ₹39.42 Cr, Viral Fever sabse upar (₹7.93 Cr)

Power BI Dashboard

Power BI me bana Major Dashboard



Major Dashboard

- Stay (days) admission status aur ward type ke hisaab se
- Treatment cost doctor aur ward type ke hisaab se
- Slicers: Ward type, age, length of stay
- Ek hi page par poora hospital overview

Jupyter Notebook Dashboard

Python (pandas + matplotlib) Major Dashboard



Major Dashboard

- Sabhi 5 files Patient ID par merge karke master dataset
- 6 KPI cards aur 9 charts ek page par
- Sab consistency checks pass hue

Key Findings aur Conclusion

Key Findings

- Total revenue **₹39.42 Cr**, avg **₹3.94 Lakh** per patient
- ICU me sirf 12.9% patients, par revenue ka **63.9%**
- Insurance sabse bada payment mode: **₹15.20 Cr (38.6%)**
- Viral Fever sabse common bimari aur sabse zyada revenue (₹7.93 Cr)
- Dr. Singh sabse zyada revenue, Dr. Verma sabse zyada patients
- 60+ age group sabse bada (32%)

Conclusion

Dashboards se hospital ka data aasani se samajh aata hai. Ward, bimari, doctor aur payment ke hisaab se kahan revenue aur load zyada hai, ye ek nazar me dikh jata hai.

Excel, Power BI aur Jupyter me alag-alag tareeke se dashboard bana kar data ko dikhaya gaya. Future me ise real hospital data se jod sakte hain.

Thank you! Questions?